The Price List

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Part One has built a world from two moves, grown numbers in it, and poured it into a vessel. This chapter reads the bills. Everything that has happened so far cost something, and the costs are not impressions: in this cosmos every price is a theorem, checked by the same silent referee as everything else in the book. The ledger of words takes its economics loans here, once, openly: price, bill, currency, cash value counted moves, counted symbols, counted heat. What the chapter finds is a price list with more than one column, columns that can disagree about which road is cheap, one item on the entire list marked free, and, at the very end, under the reader's contract, a reading that gives the whole list its oldest name.

The Schoolteacher's Rule

Generations of schoolchildren have been taught a small rule with no reason attached: when you multiply by repeated addition, put the smaller number of additions in charge. Three times one thousand, done by hand, is one thousand plus one thousand plus one thousand, three additions and done; nobody sane adds three to itself a thousand times. The rule is real economics, taught to eight year olds, and the teacher rarely says why, because in her system the why is visible on the child's slate: the count of additions is the count of labor.

Our calculus has its own slate, and on it the why is a compiled formula. The multiplication box of the number chapters, the little machine that reaches the numeral of a product, has an exact running cost: for factors m and k, the box completes in the successor of m, times k, plus two moves, every move counted, the closed form proved in the module under the name boxLen and summed over runs by rLenSum. Read the formula's shape before its digits: it is lopsided. The two factors do not sit in equal seats. The same product, seated two ways, is billed two ways, and the formula says exactly which seating is cheap, with no appeal to taste. The teacher's system had its seating rule and ours has its own, and they need not agree, because prices belong to machines, not to numbers; what the two slates share is the lesson under both, the one this chapter is about: in any world that computes step by step, order is money. There is no such thing as just multiplying. There is always a road taken, and the road has a toll, and the toll is a fact.

Notes and sources. The multiplication box and its value theorem debut in the chapter on the unreachables; the closed cost form boxLen, with the rLen family and rLenSum, is compiled in the module and reported in the Transparent Arithmetic working paper, which also states the honest boundary repeated below: these are the prices of the canonical derivation, and other strategies price differently.

The Demon's Bill

The reader met a debt collector briefly in the opening chapter. Here is the whole century he haunted. In 1867 James Clerk Maxwell, in a letter to a friend, imagined a very small, very neat being stationed at a door between two chambers of gas. The being does no work; it only watches, and sorts: fast molecules to the left, slow to the right. Given time, one chamber grows hot and the other cold, order from disorder, for free, and the second law of thermodynamics, the law that says the universe's disorder never decreases on its own, lies broken on the floor by a doorman. William Thomson gave the being its lasting name: Maxwell's demon. For a hundred years the demon haunted physics as its politest ghost, because nobody could say where the trick failed. Every exor-

cism aimed at his senses. In 1929 Leo Szilard sharpened the haunting to a single molecule and a single decision, and showed the stakes exactly: one binary fact, one bit, this side or that, is worth a fixed small ration of work, so if the demon's knowledge were free, perpetual motion would follow.

The resolution, when it finally came, is one of the most beautiful reversals in science, and the reader already holds its two halves from the opening chapter. Landauer, 1961: erasing one bit has an unavoidable heat price. Bennett, 1982: the demon's eyes are innocent. He can watch for free, sort for free, even remember for free, so long as he never throws a memory away. But he is a finite being, and his notebook fills. To keep sorting forever he must erase, and Landauer's price on the erasure pays back, exactly, to the last fraction of a joule, everything his sorting gained. In 2012 a laboratory in Lyon held a single microscopic particle and collected the debt in person, one bit, one measured puff of heat, the floor confirmed. Landauer liked to compress the century into three words, information is physical, and the slogan belongs here, in the history, where it earned its keep.

But stand back from the apparatus and look at what the demon actually is, because the chapter turns on it. He is a sorter. His entire occupation is judgment in miniature: this from that, worth keeping from worth discarding, fast from slow, a verdict per molecule, forever. And the century's verdict on him is not that judgment is impossible. It is that judgment is billed, and billed in a specific currency that nobody suspected: not in the watching, not in the deciding, but in the forgetting that continuous judgment forces. The judge's bill arrives at the wastepaper basket. Keep that sentence; the last section of this chapter will need it.

Notes and sources. Maxwell's letter to Tait, December 1867; Thomson's naming, 1874. Szilard, 'On the decrease of entropy in a thermodynamic system by the intervention of intelligent beings' (1929). Landauer (1961); Bennett's exorcism via the demon's records (1982), with his earlier reversibility result (1973); Berut and colleagues, Nature 483 (2012). The opening chapter's section on the price of forgetting carries the same physics at first meeting; this section owns the demon's full century and the judgment reading.

The Free and the Billed

Now the law of the list, and it is a theorem twice over, once in physics and once in our module. Bennett's half: computation as such has no floor. A calculation arranged so that nothing is ever discarded, every step undoable, can in principle run for free; the thinking costs nothing, the throwing away costs always. Our half, compiled: in the two-move calculus, loss has exactly one door and abundance has exactly one door, and they are different doors. The theorem kStepOcc says a K firing removes precisely the occurrences of its discarded argument, symbol for symbol, and nothing else ever removes any. The theorem sStepOcc says an S firing adds precisely the occurrences of its shared argument, and nothing else ever adds any. Every bill in this world is issued at one of two counters, and the terms that never visit the first counter, the linear world of the opening chapter, are exactly the programs a demon could run without a notebook to burn.

So the list has two currencies, and here is the finding that makes it a list rather than a column: the currencies disagree. Take the same multiplication, the same product, and seat the factors both ways. The module's meters price each road twice: in time, by the step count of boxLen, and in heat, by the duplication and erasure loads, dupMass and eraseMass, with the box's exact bills compiled as boxDup and boxErase. The result is an inversion, proved, not observed: the seating that wins the clock loses the thermometer. The fast order is the hot order. A engineer optimizing for speed and an engineer optimizing for heat will seat the same two numbers in opposite chairs, and both will be right, because within sequential form there is no costless road, only a choice of which bill to pay.

Sometimes the stakes of the choice are not even finite. At most moments in a computation several moves are legal, and the opening chapter compiled the extreme case: face a K holding a treasure in

one hand and the mirror facing the mirror in the other. The patient road drops the spinning thing unexamined, one move, done. The eager road, evaluate everything before using anything, walks into omegaCycle, the compiled circle, and never comes back. Same term, same rules, one road costs one move and the other costs everything. Arrival itself is a line item.

And then, at the bottom of the page, one entry unlike every other on it. The bill for the first theorem of the coin, A implies A, the statement that a thing is itself: its proof is three symbols, S K K, and its derivation borrows nothing. No axiom is consumed; no assumption is taken on loan; the debit column is empty, for every A there is, forever. Every other certainty on earth is purchased, from premises, from evidence, from measurement, from trust. This one is in stock before the store opens. The coin chapter, meeting the same fact from the logic side, gave it the phrasing this book will keep: the certainty that costs nothing and is always in stock. The list, then, in one look: everything billed, at two counters, in two currencies that disagree, with infinity available as a price, and one item, the self-identity of whatever is, marked free.

Notes and sources. Bennett's no-floor result for reversible computation (1973). The occurrence ledgers kStepOcc and sStepOcc, the linear fragment, and their discussion are the Linearity Ladder working paper; the meters dupMass and eraseMass with the box bills boxDup and boxErase are the Entropy Budget working paper, whose compiled inversion, the fast operand order is the hot order, this section reports; omegaCycle is the opening chapter's circle; the free item is iTyped with skIdentity, the coin papers. All prices are of the canonical derivations, as stated there.

The Portrait

One resident of this book has not yet been billed, and its account is the strangest on the list. The vessel chapter met it as a species: the term whose every collision returns itself. Whatever it encounters, the product of the meeting is one more of it, and nothing of the other survives. The chapter before priced its world; this section prices the pattern, and the pattern alone, at form level, with its name withheld for one more section.

Run its account. Income per meeting: itself, again. Not itself improved, not itself enlarged, not itself plus a durable copy; under honest bookkeeping, exactly itself, once, the meters compiled: minted, zero. Expense per meeting: four bonds of the common world, burned, the meters again: tauMealBurn, four, every cycle, paid not from its own substance but from the shared stock of structure the whole vessel lives on. Growth: none possible, by the law of the mint, since growth is minted structure and no reaction mints. Now audit the whole enterprise. Every meeting is converted entirely into maintenance. The entire activity of the pattern is the upkeep of the pattern. It produces nothing, stores nothing, contributes nothing, and burns the commons at a fixed rate to remain exactly what it was. In the one world where its accounting is never enforced, the lawless vessel where copies are free, it inherits everything, instantly, and the world ends as a monoculture of it. In every world with honest books it cannot establish a single copy, and the vessel forgets it without ever having met it.

And on the same page of the ledger, for contrast, stands the item marked free. The crown, S K K, is this pattern's exact opposite in every column: it returns not itself but the other, unchanged; its meetings cost the world nothing; its certainty is derived from nothing; and in the honest vessel it is not extinct but present, a standing citizen, remade daily by the traffic of the whole. The keeper of self burns the world and cannot grow. The returner of the other spends nothing and is always in stock. Both sentences are theorems. The reader may already suspect what the traditions call the first portrait. The last section says it, under the contract.

Notes and sources. The pattern's cycle and its fate in both vessels are the Combinator Soup working paper; the meters tauMealMint and tauMealBurn and the law of the mint, stepsNeverMint, are

the Assembly Ledger working paper; the crown's standing population is Combinator Soup, Sections 6 to 8. The withholding of the name is deliberate and ends in the next section.

The Price of the Ego

The register changes here, openly, as this book does whenever it does. Everything above is theorem and history: the formulas are compiled, the demon is documented, the meters read what they read. What follows is a reading, offered under the contract of the front note, declinable without cost, and it begins by taking the ledger's last and heaviest loan. The word is ego, and its cash value in this book is exact: the traditions' name for the self-maintaining separation pattern, the one whose form-level portrait, drawn entirely in compiled ink, is the section above.

A Course in Miracles, the teaching this book has kept company with since its first chapter, describes the ego in the language of accounting, and it does so relentlessly: sin is recorded as debt; guilt is the account book in which the record is kept; sacrifice is the ego's one law, the conviction that everything must be paid for and someone must lose; and judgment, the ego's full-time occupation, is described as a task so wearying that laying it down is called the lifting of a burden, for the exact reason the demon's century discovered: to judge everything is to sort forever, and sorting forever fills a notebook that must forever be burned. The Course's deepest economic claim is about where the expense truly lies. Truth, it says, requires no defense, no maintenance, no effort; what costs is the upkeep of separation, the ceaseless work of keeping a boundary that is not really there, and the whole tiredness of a human life is that maintenance bill. Read that claim against the chapter's ledger, entry by entry. The one item derived from nothing, kept without spending, always in stock: identity, what is, being itself. The one pattern whose entire budget is self-upkeep, which burns the commons to stay unchanged and cannot grow by its very bookkeeping: the portrait. The one activity billed not in its doing but in its forgetting: judgment, priced at the demon's wastepaper basket. And the Course's word for the end of the whole arrangement is a bookkeeping word too: Atonement, which it insists is not a payment but a cancellation, the recognition that the debt was never truly incurred. On this chapter's list that sentence has an address: the free item was always on the list, before the store opened, and every anxious purchase of certainty was the buying of something already in stock.

A Course of Love adds the frame of time, and it fits the same page. What it calls the time of learning is, in this chapter's terms, the billed economy: effortful, sequential, step-priced, every gain paid for at a counter, which is a fair description of every computation this book has run. And what it points toward, the elevated Self of form creating with means and end as one, is the free economy: no road between intention and arrival, hence no toll, the mode of the item with the empty debit column. The Course of Love's claim, in the ledger's dialect, is that a life can change which economy it is run from, and that the change is not earned, since earning is the old economy's verb, but accepted.

So the stone of this chapter, its quintessence, contracted to one line as the reader's contract requires: only forgetting is ever billed; what is kept whole costs nothing. And beside it, printed at full strength as this book's honesty demands, the deflationary reading. These are structural rhymes between a small rewriting calculus and a spiritual psychology. The calculus proves nothing about minds, nothing about egos, nothing about salvation; a skeptic will say the rhymes were purchased by our choice of words, ledger for ledger, demon for judge, and the skeptic's account balances too. The theorems hold either way, which is why they were worth a chapter; the reading is yours to keep or return; and the chair it is decided in is, as always, the reader's. What Part One can say with receipts is smaller and harder than any doctrine, and it has now said it: in a world of pure form, being what one is costs nothing, keeping oneself apart burns the world at a metered rate, and the whole price of the running universe is paid at the counter marked forgetting. The next chapter pours this priced world into a sealed vessel and watches what such an economy grows.

Notes and sources. A Course in Miracles on sacrifice as the ego's law, guilt as record, judgment as burden, truth as needing no defense, and Atonement as cancellation rather than payment, paraphrased from the Text, Workbook, and Manual; A Course of Love on the time of learning and on means and end as one, paraphrased from the Treatises and Dialogues; both under the front note's contract, with no claim that either Course teaches thermodynamics or that thermodynamics teaches either Course. The ledger anchors of the reading are the sections above: iTyped and skIdentity for the free item; tauMealMint, tauMealBurn, and stepsNeverMint for the portrait; kStepOcc for the counter of forgetting; Bennett 1982 for the judge's bill. The chapter order note: this chapter precedes the vessel in the table of contents, and its closing line hands off accordingly.